

Bonsai Federation Analysis - Autodesk User Manual

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Introduction

What is Bonsai Federation Analysis?

Bonsai is an **open-source BIM coordination tool** built on Blender and IfcOpenShell. It provides:

-  **Fast clash detection** using spatial R-tree indexing
-  **Intelligent cascade grouping** (identifies root cause elements)
-  **BCF 2.1 export** compatible with Autodesk tools
-  **Cost intelligence** for coordination decision-making
-  **Markdown + Excel reports** for stakeholder communication

Why Bonsai + Autodesk?

Complementary Strengths:

Capability	Bonsai	Navisworks	Best Use
Clash Detection	✓ Fast (R-tree)	✓ Standard	Bonsai for batch, Navis for review
Cascade Analysis	✓ AI grouping	✗ Manual	Bonsai finds root causes
BCF Export	✓ BCF 2.1	✓ BCF 2.1	Round-trip coordination
Cost Intelligence	✓ Integrated	✗ No	Bonsai for business case
3D Navigation	⚠ Blender UI	✓ Optimized	Navis for daily review
License Cost	🆓 Free	💰 \$3,600/year	Bonsai for tight budgets

Recommended Workflow: 1. **Bonsai:** Batch clash detection + intelligent analysis 2. **Export BCF:** Share findings with Autodesk teams 3. **Navisworks:** Review in familiar interface 4. **Update BCF:** Mark status, add comments 5. **Round-trip:** Import updated BCF into Bonsai for verification

Quick Start Guide

For Navisworks Users Receiving BCF from Bonsai

Step 1: Receive BCF File

You receive: ClashReport_Week12.bcfzip (email or cloud)

Step 2: Import into Navisworks 1. Open Navisworks Manage 2. **Viewpoint** tab → **BCF** → **Import**
3. Select `ClashReport_Week12.bcfzip` 4. Click **Open**

Step 3: Navigate to Clashes 1. **Viewpoint** panel → Expand topics 2. Click any topic (e.g., "Clash: Pipe vs Duct") 3. Viewport jumps to clash location 4. Red highlighting shows clashing elements

Step 4: Update Status 1. Right-click topic → **Add Comment** 2. Type: "Duct rerouted 800mm north, resolved" 3. Change status: **Open** → **Resolved** 4. **Viewpoint** → **BCF** → **Export** 5. Save as `ClashReport_Week12_UPDATED.bcfzip` 6. Email back to coordinator

Workflow: Navisworks to Bonsai

Scenario: You detect clashes in Navisworks, want Bonsai's intelligent analysis

Step 1: Export IFC Models from Navisworks

Navisworks doesn't export IFC directly. You need source models:

Option A: From Revit

Revit → File → Export → IFC
- IFC Version: IFC4 (or IFC2x3)
- Save to shared folder

Option B: From Existing IFC

If your federated model came from IFC files, use those originals

Step 2: Load IFC into Bonsai

1. **Open Blender** (with Bonsai addon installed)
2. **Bonsai** menu → **Federation** panel
3. Click "**Load Federation Database**"
4. Select your federation database (or create new)
5. IFC models auto-loaded into spatial database

Step 3: Run Clash Detection in Bonsai

1. **Federation** panel → **Clash Detection** section
2. Select disciplines:
3. **Discipline A**: MEP (e.g., ACMV, Plumbing)
4. **Discipline B**: Structure
5. Click "**Detect Clashes (Preview)**"

6. Results appear in panel (e.g., "252 clashes found")

Step 4: Generate Intelligent Clash Report

1. Click "**Generate Clash Report**"
2. Bonsai analyzes clashes:
3.  Groups cascading clashes (finds root causes)
4.  Calculates resolution costs
5.  Prioritizes by severity
6. Report saved to `clash_reports/clash_resolution_YYYYMMDD_HHMMSS/`

Step 5: Review Report (Markdown Format)

 `report.md` - Open in any text editor or Markdown viewer
 `snapshots/` - PNG images of each cascade group

Key Sections: - **Executive Summary:** Total clashes, cascade efficiency - **Group 1: CASCADE CLASH PATTERN** - Root cause: "Opening (2eD_GMhN)" - Affected elements: 8 clashes - Resolution strategy: "Adjust opening position" - Estimated cost: \$1,200 design + \$800 construction

Step 6: Export BCF for Navisworks Team

1. **Federation** panel → **BCF Export**
2. Click "**Export BCF 2.1**"
3. Select clashes to include (or select all)
4. BCF file generated: `clash_export_YYYYMMDD.bcfzip`
5. Share with Navisworks users

Workflow: Bonsai to Navisworks

Scenario: Bonsai detects clashes, Navisworks team resolves them

Step 1: Bonsai Generates BCF

(As per Step 6 above)

Step 2: Navisworks Imports BCF

Navisworks Manage:

1. Viewpoint → BCF → Import
2. Select clash_export_YYYYMMDD.bcfzip
3. Topics appear in Viewpoint panel

Step 3: Navisworks User Reviews Clashes

What You See: - Topic: "Clash: Pipe Fitting (Aras 02) vs Opening" - Snapshot: 3D view with red highlighting - Description: Element details, disciplines, status - Comments: Any notes from Bonsai coordinator

What You Do: 1. Click topic → Viewport jumps to clash 2. Use Section Box to inspect closely 3. Measure clearances with Measure Tools 4. Decide resolution: - **Reroute:** Change MEP layout - **Adjust:** Move opening/structural element - **Acceptable:** Document why (e.g., "Non-critical, 50mm clearance OK")

Step 4: Update BCF Status

Right-click topic:

- Add Comment: "Pipe rerouted via north corridor, clearance 200mm"
- Change Status: Open → Resolved
- Assign To: [Your name]

Step 5: Export Updated BCF

Viewpoint → BCF → Export

- Filename: clash_export_YYYYMMDD_RESOLVED.bcfzip
- Send back to Bonsai coordinator

Step 6: Bonsai Verifies Resolutions

(Coordinator imports updated BCF into Bonsai, re-runs detection to confirm)

BCF Export & Import

BCF 2.1 Compatibility

Bonsai exports **BCF 2.1** format, fully compatible with:

Software	Import	Export	Round-trip
Navisworks 2024	✓	✓	✓
Navisworks 2023	✓	✓	✓
Revit 2024	✓	✓	✓
BIM 360 / ACC	✓	✓	✓
Solibri	✓	✓	✓
BIMcollab	✓	✓	✓

What's Included in Bonsai BCF?

Per Topic (Clash):

- ✓ Topic GUID (unique ID)
- ✓ Title: "Clash: Pipe Fitting vs Opening"
- ✓ Description: Element details, disciplines
- ✓ Status: Open / In Progress / Resolved
- ✓ Priority: High / Medium / Low
- ✓ Viewpoint:
 - Camera position (location, target, up vector)
 - Element GUIDs (IFC references)
- ✓ Snapshot: PNG image (800x600, ~200KB)
 - Red highlighting on clashing elements
 - Isometric view (~45° angle)

BCF Import into Navisworks

Automatic Mapping:

Bonsai BCF Field	Navisworks Field
Topic Title	Issue Name
Description	Comments
Status (NEW)	Active
Status (RESOLVED)	Closed
Priority (HIGH)	Critical
Viewpoint → Camera	Saved Viewpoint
Element GUIDs	Search Sets (if IFC loaded)

Known Limitations:

 **GUID Matching:** Navisworks must have **same IFC files loaded** for element highlighting to work - If models differ → Viewpoint works, but element selection may fail - Solution: Ensure Navisworks uses **exact same IFC exports** as Bonsai

 **Snapshot Rendering:** Bonsai snapshots are Blender-rendered - May look different from Navisworks viewport - Viewpoint data is accurate, visual style differs

BCF Import into Revit

Steps:

Revit 2024:

1. Add-Ins → BCF Manager
2. Import BCF
3. Issues appear in BCF Browser
4. Click issue → Model view aligns to viewpoint

Limitation: - Revit BCF import requires **model elements** in current file - If clash involves linked models → Manual navigation needed

Clash Reports for Autodesk Teams

Report Format Options

Bonsai generates **Markdown (.md) reports** optimized for Autodesk workflows:

Option 1: View in Text Editor

- Notepad++ (Windows)
- VS Code (Windows/Mac/Linux)
- Any Markdown viewer

Option 2: Convert to PDF

```
# Using Pandoc (free tool)
pandoc report.md -o report.pdf
```

Option 3: Convert to Word

```
pandoc report.md -o report.docx
```

Option 4: View in Browser

- Drag report.md into Chrome/Edge
- Install Markdown Viewer extension

Report Structure

Executive Summary (Page 1)

- Total clashes: 252
- Cascade groups: 13 (62 clashes grouped)
- Ungrouped: 190 isolated clashes
- Estimated cost: \$26,169 (design + construction)
- ROI: 1.2x (every \$1 in design saves \$1.20 in field)

Per Cascade Group (Pages 2+)

```
## GROUP 1: CASCADE CLASH PATTERN

Root Cause Element: Opening (GUID: 2eD_GMhN)
Impact: 8 clashes
Severity: HIGH

Affected Neighboring Elements:
| # | Element A | Element B | Disciplines | Status |
|---|-----|-----|-----|-----|
| 1 | [Pipe Fitting (Aras 02)](snapshots/group_1_overview.png) | [Opening (Aras 02)](snapshots/group_1_overview.png) | [Plumbing] | [Open] |
| 2 | [Pipe Segment (Aras 02)](snapshots/group_1_overview.png) | [Opening (Aras 02)](snapshots/group_1_overview.png) | [Plumbing] | [Open] |

[Snapshot: Red highlighted elements in 3D view]

Resolution Strategy:
- Option 1: Adjust opening position (Cost: $400)
- Option 2: Reroute plumbing (~$1,200)

Recommendation: Adjust opening (cheaper, less cascade impact)
```

Using Reports in Coordination Meetings

Typical Agenda:

1. Present Executive Summary (5 min)
2. Show stakeholders total clashes, cost impact
3. Highlight cascade groups vs isolated issues

4. Review High-Priority Groups (20 min)

5. Show snapshots on screen
6. Discuss root causes
7. Assign responsibility per discipline

8. Assign Actions (10 min)

9. MEP: Resolve Groups 1-5 (25 clashes)
10. Structure: Resolve Groups 6-8 (15 clashes)
11. Architecture: Adjust openings in Groups 1, 3
12. **Set Deadlines**
13. Critical: Resolve by Friday
14. Major: Resolve by next week
15. Normal: Ongoing

Pro Tip: Print PDF report and distribute to all attendees

Troubleshooting

"BCF Import Failed in Navisworks"

Symptoms:

Error: "Cannot import BCF file"

Solutions:

1. **Check BCF Version**
2. Bonsai exports BCF 2.1
3. Navisworks 2021+ supports BCF 2.1
4. Older Navisworks → Upgrade or use Solibri
5. **Verify ZIP Structure** clash_export.bcfzip |— bcf.version (XML) |— markup.bcf (XML) |— [topic-guid]/ |— markup.bcf |— snapshot.png

6. Try Manual Extraction

7. Rename `.bcfzip` → `.zip`
8. Extract contents
9. Import individual markup.bcf files

"Viewpoints Don't Match in Navisworks"

Symptoms: - BCF imports successfully - Clicking topic → Viewport doesn't move correctly

Cause: Coordinate system mismatch

Solution: 1. Verify Navisworks units match IFC units (meters) 2. Check if federated model has correct origin 3. Re-export IFC with consistent coordinates

"Element GUIDs Not Found"

Symptoms: - BCF imports - Viewpoint works - But elements not highlighted/selected

Cause: Navisworks doesn't have matching IFC elements

Solution: 1. **Check IFC is Loaded** - Navisworks → Selection Tree → Verify IFC files present 2. **Verify GUID Matching** - Bonsai uses IFC GUIDs (GlobalId) - Navisworks must have **same IFC files** Bonsai used 3. **Reload Models** - File → Append → Add IFC files - Refresh Selection Tree

"Snapshots Look Different"

Expected Behavior: Bonsai snapshots are Blender-rendered, Navisworks uses different rendering engine

Not a Bug: Visual style differs, but viewpoint data is accurate

If Concerned: - Use Viewpoint data (camera position) as source of truth - Snapshots are reference only

FAQ

Can Bonsai replace Navisworks?

Short Answer: Not entirely, but it can **reduce Navisworks seats** significantly.

Long Answer: - **Batch Clash Detection:**  Bonsai is faster (R-tree indexing) - **Intelligent Analysis:**  Bonsai's cascade grouping > Navisworks manual grouping - **Daily Review:**  Navisworks has better 3D navigation for end users - **Coordination Meetings:**  Either works (preference-based)

Recommended Split: - **1 Bonsai seat** (coordinator): Batch detection + analysis - **2-3 Navisworks seats** (team leads): Daily review + meetings - **Everyone else:** BCF import into Revit/Solibri/free viewers

Savings: \$7,200/year (avoiding 2 Navisworks licenses)

Does Bonsai support BCF 3.0?

Currently: BCF 2.1 only

Reason: BCF 2.1 is industry standard (2014), widely supported

BCF 3.0: Newer (2019), limited adoption (Solibri, BIMcollab)

Autodesk Support: - Navisworks 2024: BCF 2.1 , BCF 3.0  - BIM 360: BCF 2.1 , BCF 3.0  (partial)

Roadmap: BCF 3.0 support planned for Bonsai v2.0 (2026)

Can I edit clash reports?

Yes! Reports are **editable Markdown (.md) files**

Common Edits:

1. Add project-specific notes
2. Customize resolution strategies
3. Update cost estimates
4. Add meeting action items

Workflow: 1. Open `report.md` in text editor (VS Code, Notepad++) 2. Edit sections (Markdown syntax) 3. Save file 4. Convert to PDF: `pandoc report.md -o report.pdf` 5. Distribute to team

What if clashes are resolved in Revit, not Navisworks?

Workflow:

1. **Receive BCF from Bonsai** → `clashes.bcfzip`
2. **Import into Revit:**
3. Add-Ins → BCF Manager → Import
4. **Resolve Clashes in Revit:**
5. Modify elements directly
6. Update BCF status in BCF Browser
7. **Export Updated BCF:**
8. BCF Manager → Export
9. **Send to Coordinator:**
10. Email `clashes_RESOLVED.bcfzip`

Bonsai Verification: - Coordinator imports updated BCF - Re-exports IFC from Revit - Re-runs clash detection to verify

How do I share reports with non-technical stakeholders?

Problem: Stakeholders don't use BIM software

Solution: Convert to Familiar Formats

Option 1: PDF Report

```
pandoc report.md -o clash_report.pdf
```

-  Universal format (print, email, archive)
-  Embeds snapshots automatically
-  Readable by anyone

Option 2: PowerPoint (for meetings)

```
pandoc report.md -o presentation.pptx
```

-  Edit slides in PowerPoint
-  Add company branding
-  Present in coordination meetings

Option 3: Excel (for tracking) - Export clash table to CSV - Import into Excel for status tracking - Add formulas for % resolved

Option 4: Email Summary

Subject: Clash Detection Summary - Week 12

Executive Summary:

- 252 clashes detected
- 13 cascade groups identified
- Estimated impact: \$26,000

Action Required:

- MEP Team: Resolve 62 clashes (Groups 1-5)
- Structure: Resolve 15 clashes (Groups 6-8)

Deadline: Friday, 5 PM

Full report attached (PDF)

Additional Resources

Bonsai Documentation

- **Official Docs:** <https://docs.bonsaibim.org> (*hypothetical*)
- **GitHub Repo:** <https://github.com/lfcOpenShell/lfcOpenShell>
- **Community Forum:** BlenderBIM Add-on discussions

BCF Resources

- **buildingSMART BCF Spec:** <https://www.buildingsmart.org/standards/bsi-standards/bim-collaboration-format-bcf/>

- **BCF Viewers:** BIMcollab Zoom (free), Solibri Anywhere (free)

Autodesk Resources

- **Navisworks BCF Guide:** Autodesk Knowledge Network
 - **Revit BCF Manager:** Add-Ins documentation
 - **BIM 360 BCF Workflow:** Autodesk Construction Cloud docs
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Support & Contact

For Bonsai Issues: - GitHub Issues: [lfcOpenShell/lfcOpenShell/issues](https://github.com/lfcOpenShell/lfcOpenShell/issues) - Community Forum: OSArch.org

For BCF Compatibility Issues: - Email: [your-support-email] - Include: BCF file, Navisworks version, error message

For Training: - Contact BIM coordination team for onboarding sessions

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